

Lesson: Diffused RGB LED

Big Picture

This lesson will introduce the diffused RGB (red-green-blue) LED while allowing students to become more acquainted with the BBC micro:bit microcontroller hardware and Makecode *Blocks Editor* software tool. The students will create a program that will vary the color of the RGB LED using the A and B buttons on the micro:bit.

Objectives

Students will be able to:

- Define Variable
- Define Selection (decision)

Alabama Standards Alignment

7 (Fifth Grade): Identify Variables.

- ▮ Examples: Determine if a variable is required for use later in the program.

8 (Fifth Grade): Demonstrate the programs require known starting values that may need to be updated appropriately during the execution of programs

- ▮ Examples: create a program that sets a variable to an initial value then later updates (changes) the value of the variable.

7: Create algorithms.

- ▮ Examples: Create algorithms that demonstrate sequencing, selection, or iteration.

Links to Resources

XXX How a pushbutton works on the inside: <https://www.youtube.com/watch?v=vskNLQZp6Xg>

Preparation

- deiffused_rgb_led_student_handout: Tutorial handout found on lesson page

Choose a presentation method:

- Instructor may lead the students through the lesson using the student handout
- Students may work at their own pace using the student handout. You may also post the video and tutorial locally and allow students to choose.

Materials Required

Each student (or pair of students) requires:

- Tutorial handout
- micro:bit
- USB cable
- diffused RGB LED
- 100 Ω resistor (3)

- Jumper wires (4)
- Breakout
- MakeCode editor
- Internet connected computer with modern browser

**Note: Browsers known to work with micro:bit software includes Firefox, Chrome, Safari, and Microsoft Edge
For a complete list, visit this page: <https://makecode.microbit.org/browsers>*

Vocabulary and Concepts

Variable: A named container for storing a value that may change while the program is running.

Selection: Using a boolean condition to determine which of two parts of an algorithm is used.

Teaching Guide

Getting started (10 mins)

Tell the class that they will create a micro:bit program with a RGB LED on a breadboard today. Before they start programming, everyone needs to learn the new vocabulary terms.

Activity (40 mins)

The class is now ready to create their micro:bit with the RGB LED. Use your chosen method to demonstrate how to complete the activity. Make sure students are able to witness the LED change colors when the various buttons are clicked. Once students get the LED to change colors, allow students to experiment by changing the color increment values. Encourage students to share their results with the class. It is important to build a sense of accomplishment early in CS Making so that students will be engaged quickly and are more likely to persevere when projects become more challenging.

Wrap Up (5 mins)

Review the 2 - vocabulary words.

Variable: A named container for storing a value that may change while the program is running.

Selection: Using a boolean condition to determine which of two parts of an algorithm is used.